www.csvtuonline.com

324611 (24)

BE (6th Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical

POWER SYSTEM ANALYSIS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) is compulsory and attempt any two from remaining parts.

Q. 1. (a) Define short circuit capacity. 2

(b) Show that complex power can be computed from symmetrical components of voltage &

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

current of an unbalanced three phase system. 7

(c) Write a technical note on phase shift in Star-Delta transformer bank. Draw neat vector diagrams. 7

(d) The estimated short circuit MVA at the bus-bars of a generating station A is 1500 MVA and of another station B is 1200 MVA. The generated voltage at each station is 33 KV. If these stations are interconnected through a line having a reactance of 1 ohm and negligible resistance. Calculate the possible short circuit MVA at both station. 7

www.csvtuonline.com

www.csvtuonline.com

Q. 2 (a) Define unsymmetrical faults in power system. 2

(b) Show that for a double line to ground fault zero sequence components of currents is zero. Hence find the value of other sequence components. 7

(c) A 50 MVA, 11 KV three phase alternator was subjected to different types of faults. The fault currents are : three phase fault = 2000 A, Line to line fault = 2600 A, Line to ground fault = 4200 A. The generator neutral is solidly grounded. Find the values of three sequence reactances of alternator. Ignore resistance. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(d) A 20000 KVA, 13.8 KV generator has a direct axis subtransient reactance of 0.25 per unit. The negative and zero sequence reactances are 0.35 and 0.10 per unit respectively. The neutral of the generator is solidly grounded. Determine the subtransient current in the generator and the line to line voltages for sub transient condition when a single line to ground fault occurs at the generator terminals with the generators operating unloaded at rated voltage. Neglect resistance. 7

Q. 3. (a) Define stability of a power system and write its types. 2

www.csvtuonline.com

www.csvtuonline.com

(b) For a certain power factor load the sending end and receiving end voltages of a short transmission line of impedance (R + jX) are equal. Determine the ratio of X / R so that maximum power is transmitted over the line. 7

(c) Explain equal area criterion for stability of a power system. Draw neat diagrams. 7

(d) The steady state limit of a power system is 150 MW. A generator with constant excitation is supplying 75 MW to the system. Estimate the maximum permissible sudden increase in generator output without causing instability. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Q. 4. (a) Write three types of buses with specified quantities. 2

(b) Show that diagonal elements of a Y_{bus} is equal to the sum of admittance directly connected to that bus and an off diagonal element is equal to the negative of the sum of admittance directly connected between the buses. 7

(c) Draw flow chart of Newton - Raphson method for load flow studies including PV busses. Explain. 7

www.csvtuonline.com

www.csvtuonline.com

(d) Three busses are interconnected. Determine Y_{bus} of this system if line series impedances are :

Line (bus to bus)	Impedances (p.u.)
1 – 2	0.06 + j0.18
1 – 3	0.03 + j0.09
2 – 3	0.08 + j0.24

Neglect shunt capacitance of lines. 7

Q. 5. (a) Write indicators for active power and reactive power in a power system. 2

(b) Draw the block diagram of alternator voltage regulator scheme. Explain the function of each block. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) Explain reasons for limits on frequency in a power system. 7

(d) A 100 MVA synchronous generator operates on full load at a frequency of 50 Hz. The load is suddenly reduced to 50 MW. Due to time lag in governor system, the steam valve begins to close after 0.4 sec. Determine the change in frequency that occur in this time. H = 5 kW-sec / KVA of generator capacity. 7

www.csvtuonline.com

www.csvtuonline.com